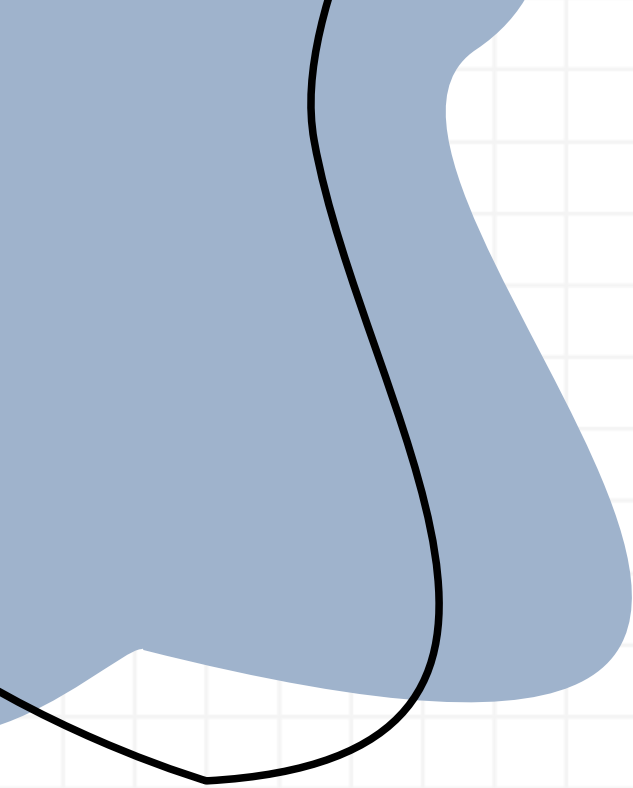




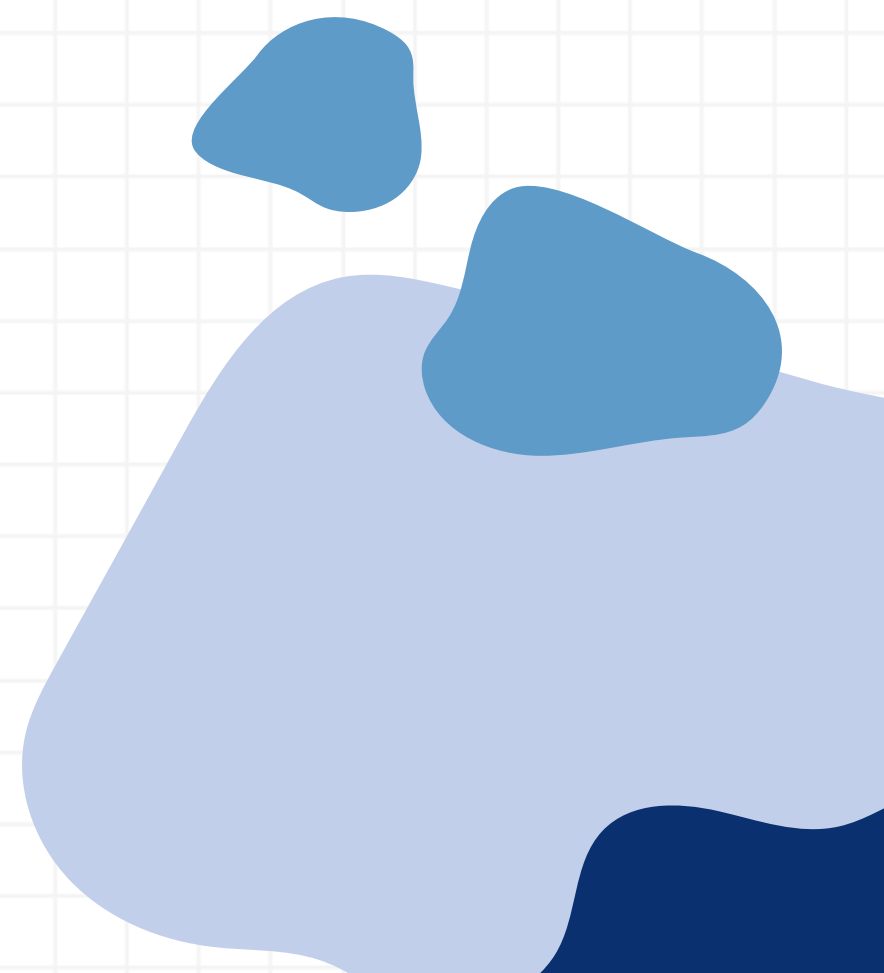
Nonlinear Portfolio Optimization Mean-Variance and Sharpe Ratio Maximization under Budget and Long-Only Constraints

Using SLSQP and local-optimum escape strategies



Group 11

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Why portfolio optimization?



How should capital be allocated across assets?

Problem definition

Portfolio Weight Vector:

$$w = (w_1, \dots, w_n)^T$$

Expected Return Vector:

$$\mu = (\mu_1, \mu_2, \dots, \mu_n)^T$$

Covariance Matrix:

$$\Sigma$$

Portfolio Expected Return:

$$R_p = w^T \mu$$

Portfolio Risk (Standard Deviation):

$$\sigma_p = \sqrt{w^T \Sigma w}$$

Budget Constraint:

$$\sum_{i=1}^n w_i = 1$$

Long-Only Constraint:

$$w_i \geq 0$$

Sharpe ratio

Optimization Goal:
maximize Sharpe
Implemented Objective:
minimize (-Sharpe)

$$S(w) = \frac{w^T \mu - r_f}{\sqrt{w^T \Sigma w}}$$

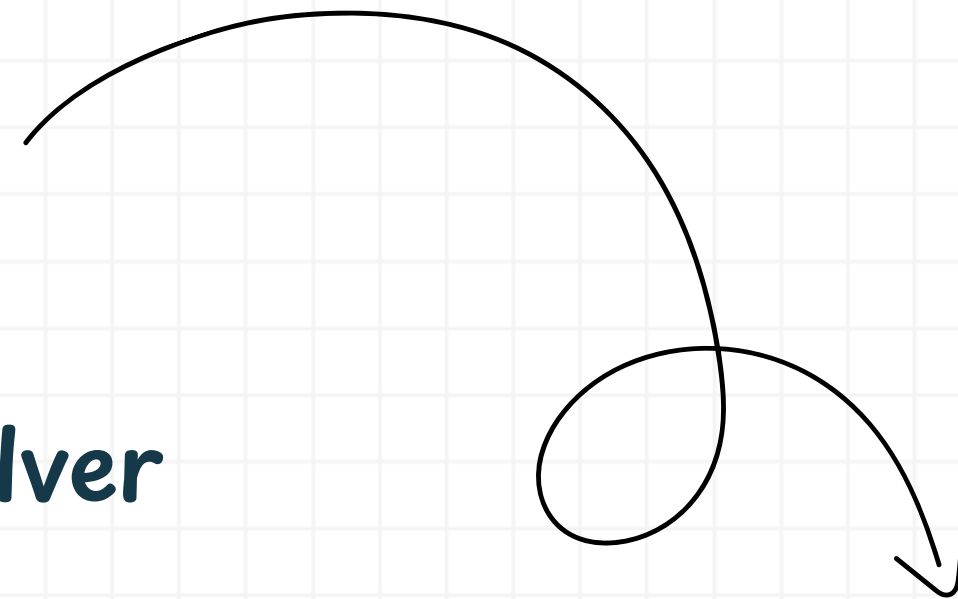
excess return

risk

We maximize Sharpe ratio
because it measures risk-adjusted
performance. Since SLSQP in
SciPy minimizes, we minimize the
negative Sharpe ratio.

Why SLSQP?

- nonlinear objective
- equality constraints
- inequality constraints
- bounds on variables
- practical numerical solver



**Sequential Least Squares Quadratic
Programming**

How SLSQP Works

Nonlinear Problem



Local Quadratic Approximation



Quadratic Programming Subproblem



Update Solution



Repeat Until Convergence

How SLSQP Works

General constrained problem:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & h(x) = 0 \\ & g(x) \leq 0 \\ & x_L \leq x \leq x_U \end{aligned}$$

Newton-style iteration:

$$\begin{aligned} y_{k+1} &= y_k + \Delta y_k \\ F'(y_k) \Delta y_k &= -F(y_k) \end{aligned}$$

QP form:

$$\begin{aligned} \min_d \quad & \frac{1}{2} d^T B_k d + \nabla f(x_k)^T d \\ \text{s.t.} \quad & A_k d = -c_k \end{aligned}$$

Lagrange function of problem:

$$\mathcal{L}(x, \lambda) = f(x) - \lambda^T h(x)$$

Lagrange optimality conditions:

$$\begin{aligned} \nabla_x \mathcal{L}(x, \lambda) &= \nabla f(x) - A(x)^T \lambda = 0, \\ h(x) &= 0. \end{aligned}$$

- This is a system of $n + m_e$ equations in (x, λ)

$$A(x) = \nabla h(x) \in \mathbb{R}^{m_e \times n}$$

Newton implementation:

$$\text{Define } F(x, \lambda) = \begin{bmatrix} \nabla f(x) - A(x)^T \lambda \\ h(x) \end{bmatrix}$$

$$\text{The Newton step } \begin{bmatrix} \Delta x \\ \Delta \lambda \end{bmatrix} \text{ satisfies: } F'(x_k, \lambda_k) \begin{bmatrix} \Delta x_k \\ \Delta \lambda_k \end{bmatrix} = -F(x_k, \lambda_k)$$

The Jacobian F' is the KKT matrix :

$$\begin{bmatrix} \nabla_{xx}^2 \mathcal{L}(x_k, \lambda_k) & -A(x_k)^T \\ A(x_k) & 0 \end{bmatrix} \begin{bmatrix} \Delta x_k \\ \Delta \lambda_k \end{bmatrix} = - \begin{bmatrix} \nabla f(x_k) - A(x_k)^T \lambda_k \\ h(x_k) \end{bmatrix}$$

Local optimum concern

nonlinear problems may
have local optima

Why test multiple starting strategies?

we tested several
starting strategies

solver output may
depend on initialization

Local optimum escape strategies

Multi-Start SLSQP

- Run SLSQP from many random feasible starting points and keep the best result.

Smart Initialization

- Start from carefully selected portfolios.

Hybrid PSO + SLSQP

- Use PSO for global exploration, then SLSQP for local refinement.

experimental observation

What happened in practice?

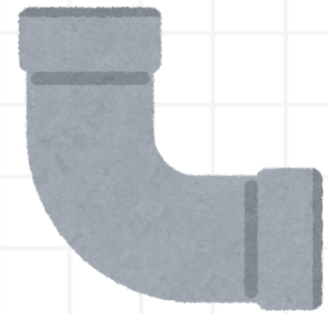
- different methods were tested
- output portfolios were very similar
- Sharpe ratios were close
- final weights were close

Reducing Sharpe Maximization to QP:

$$\begin{array}{ll} \min_w & \frac{1}{2} w^\top \Sigma w \\ \text{subject to} & w^\top (\mu - R_f) = 1, \end{array}$$

Data preparation

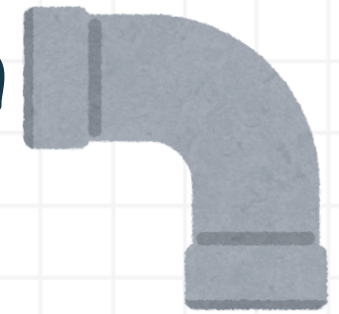
Raw adjusted data



Weekly aggregation



Returns calculation

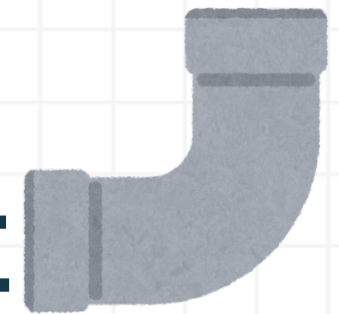


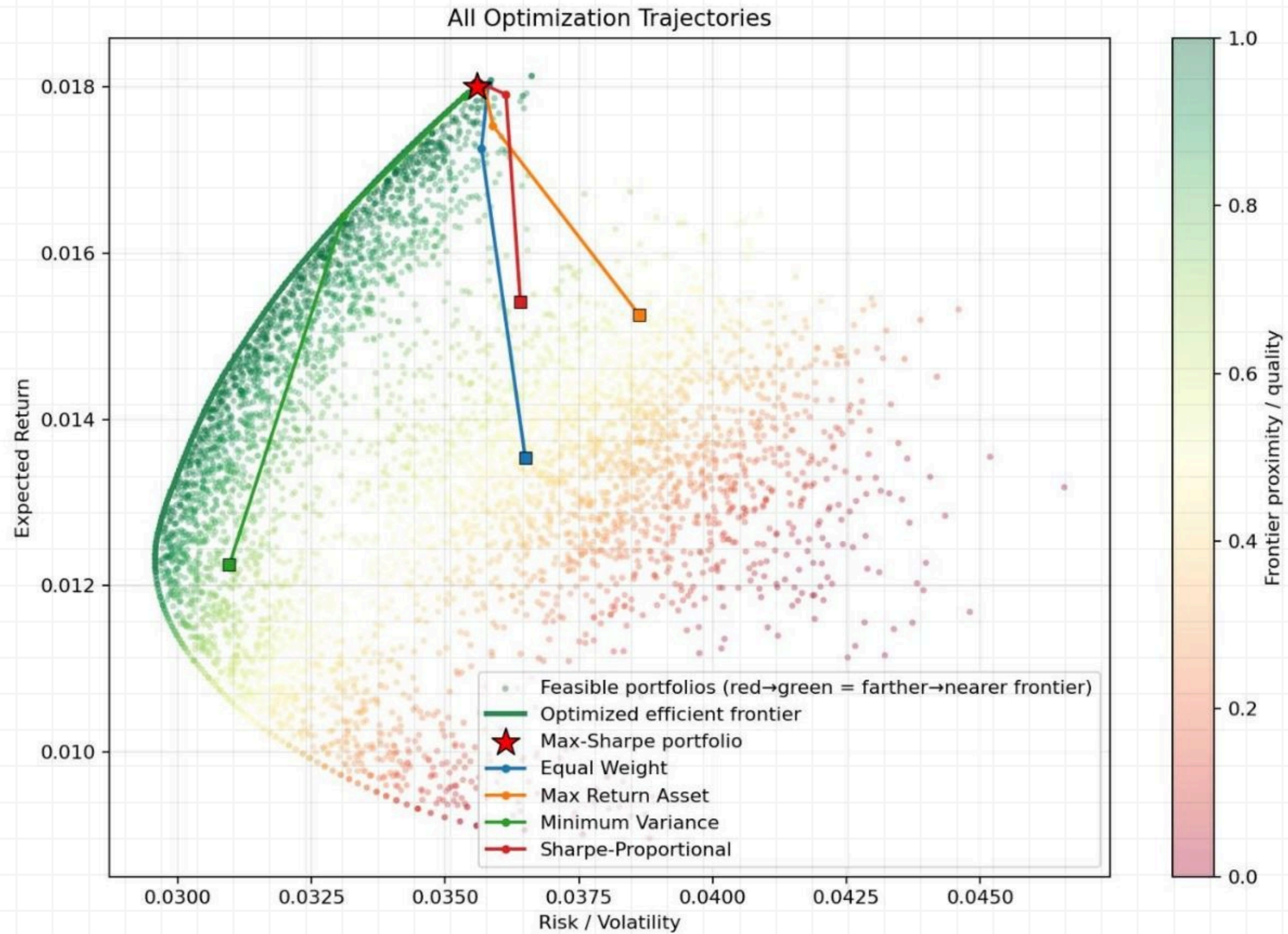
Expected returns μ

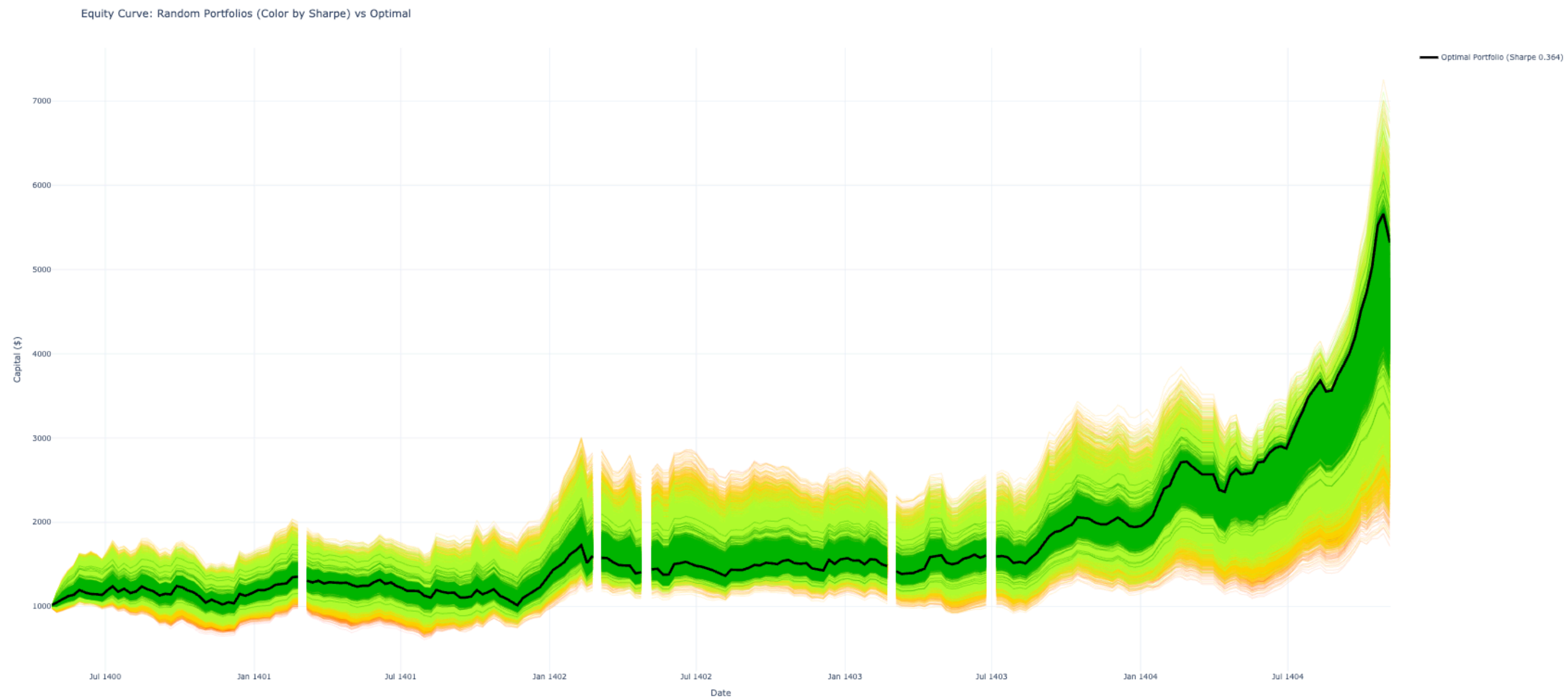
Optimization



Covariance matrix Σ









Final Optimal (Equal Weight)
Final Optimal (Max Return Asset)
Final Optimal (Minimum Variance)
Final Optimal (Sharpe-Proportional)

Thank you